



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Mechanical Engineering
Summer Term Exam
Academic Year: 2024-25 (Semester -II)

Class: TY B. Tech.	Subject: Computational Fluid Dynamics	Subject Code: PEC-ME321
Date: 23.06.2025	Time: 10:00 to 13:00	Max. Marks: 100

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. This paper contains two pages.

Q.N.		Marks	CO	BT
Q.1	Given the function $f(x) = \sin(x)$, determine the first derivative of $f(x)$ at $x = \pi/4$ using forward, backward and central differencing approximation. Compare the results with the exact analytical value. Use a step size of $\Delta x = \pi/8$ and find out the % of error for each case with respect to the exact solution.	10	CO2-3	BT1-3
Q2	Determine the order, linearity, and nature (elliptic, hyperbolic or parabolic) of the following PDE: $2 \frac{\partial^2 T}{\partial x^2} + 4 \frac{\partial^2 T}{\partial x \partial y} + 3 \frac{\partial^2 T}{\partial y^2} = 0$ Where, α and β are positive, real constants.	10	CO 2	BT3
Q3	For 1-D steady state heat conduction system without heat generation, consider a large plate of thickness $L=1$ cm with constant thermal conductivity $k=5$ W/mK. The faces are at (Left face) 100°C and (Right face) 500°C . Assuming conduction only along the thickness, arrive at the discretised equations for temperature, considering 5 grid points. Solve the matrix with Gauss elimination method and compare with the analytical solution. 	15	CO1-4	BT2-4
Q4	Explain each step involved in the SIMPLE algorithm and write the corresponding pressure correction equation.	15	CO3-4	BT1-4
Q5	a. Consider 1D wave equation, $\frac{\partial u}{\partial t} + c \frac{\partial u}{\partial x} = 0$ Find out the stability analysis of above 1D wave equation using Forward	10	CO2-3	BT2-4

		in Time and Central in Space (FTCS) discretization.			
	b.	Explain and derive that how Lax method stabilize the above 1D wave equation using Forward in Time and Central in Space (FTCS) discretization.	10		
Q6		Discuss the purpose of using a staggered grid in the Primitive Variable Approach. Also, explain one key advantage of employing a staggered grid.	10	CO2-3	BT1-4
Q7		Explain the physical significance of the Courant–Friedrichs–Lewy (CFL) number in numerical simulations.	05	CO2-3	BT1-4
Q8		What is the advantage of the Primitive Variable Approach over the Stream Function-Vorticity?	05	CO2-3	BT1-4
Q9		What are the approaches to discretizing the 1D unsteady heat conduction equation, utilizing the explicit, implicit, and Crank-Nicolson methods? Also, could you explain the advantages and disadvantages of each of these methods?	10	CO2-3	BT3

Bloom's Taxonomy (BT):: BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Express numerical modeling and its role in the field of fluid flow and heat transfer.
CO 2	Apply the various discretization methods, solution procedures and turbulence modeling to solve fluid flow and heat transfer problems.
CO 3	Interpret the knowledge, capability of analyzing and solving any concept or problem associated with heat energy & fluid dynamics.
CO 4	Use a CFD tool effectively for practical problems leading to enhance employability.

	Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded Department of Mechanical Engineering Summer Term Examination Academic Year: 2024-25 (Semester – I)	
Class: TY	Subject: Internal Combustion Engines and E-Vehicles	Subject Code: PEC-ME-311
Date: 21.6.2025	Time: 10:00 to 13:00	Max. Marks: 100

Note:

- Assume suitable data whenever necessary and state it clearly.
- The symbols carry the usual meaning.
- This paper contains 2 pages

Q.N.		Marks	CO	BT
Q.1	a The mechanical efficiency of single cylinder four stroke engine is 80%. The frictional power is estimated to be 25 kW. Calculate the indicated power and break power developed by the engine	8	CO1	2
	b An engine working on otto cycle has a cylinder diameter of 200 mm and stroke 250 mm. The clearance volume is 1570 cc. Find the air-standard efficiency. Assume $C_p = 1.004 \text{ KJ/kgK}$ and $C_v = 0.717 \text{ KJ/kgK}$ for air.	7	CO1	3
Q.2	a With neat sketch explain working of four stroke SI engine	8	CO1	2
	b With neat sketch explain working of two stroke engine	7	CO1	2
Q.3	a Explain functioning of carburettor with neat sketch.	5	CO2	2
	b Explain stages of combustion in CI engine with neat sketches	5	CO2	2
	c Explain the phenomenon of knock in CI engine and compare it with SI engine knock.	5	CO2	3
Q.4	a Why Morse test is not suitable for single cylinder engine? Describe the method of finding friction power sing Morse test.	5	CO3	3
	b Explain various exhaust and non-exhaust emission from the IC engine	5	CO3	2
Q.5	During a trial of a single cylinder petrol engine and the following observations were made: Compression ratio = 15, Oil consumption 10.2 kg/h, Calorific value of fuel= 43890 kJ/kg, Air consumption = 3.8 kg/min Speed = 1900 rpm. Torque on the brake drum = 186 N-m Quantity of cooling water used = 15.5kg/min, Temperature rise of cooling water = 36°C, Exhaust gas	15	CO2	4

		temperature = 410°C, Room temperature = 20°C, C_p for exhaust gases 1.17 kJ/kg K Calculate: (i) Brake power, (ii) Brake specific fuel consumption, and (iii) Brake thermal efficiency (iv) Also draw heat balance sheet on minute basis.			
Q.6		In a test of a 4-cylinder, 4 stroke engine 75mm bore and 100 mm stroke, the following results were obtained at full throttle at a particular constant speed and with fixed setting of fuel supply of 6.0 kg/h. B.P. with all cylinders working = 15.6 kW B.P. with cylinder no. 1 cut out = 11.1 kW B.P. with cylinder no. 2 cut out = 11.03 kW B.P. with cylinder no. 3 cut out = 10.88 kW B.P. with cylinder no. 4 cut out = 10.66 kW If the calorific value of the fuel is 83600 kJ/kg and clearance volume is 0.0001 m ³ , Calculate: (i) Mechanical efficiency, (ii) Indicated thermal efficiency, and (iii) Air standard efficiency.	15	CO2	4
Q.7	a	What are the key components of electric vehicle. Explain in brief.	5	CO4	BT2
	b	Explain in brief: Hybrid electric vehicle with general layout	5	CO4	BT2
	c	Explain the working of Lithium-ion battery with neat sketch	5	CO4	BT2

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Exhibit fundamentals of internal combustion engines.
CO 2	Demonstrate working and construction of internal combustion engines.
CO 3	Show competency about combustion in SI and CI Engines along with emission of pollutants and related control measures.
CO 4	Exhibit fundamentals of Modern Electric and Hybrid Electric Vehicles

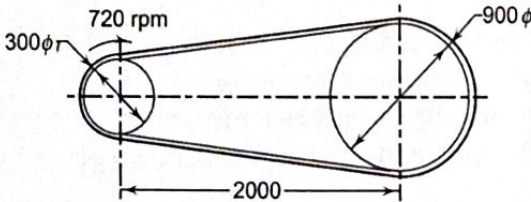


SGGS Institute of Engineering and Technology, Vishnupuri, Nanded.
Department of Mechanical Engineering
Academic Year: 2024-2025 [SEM. – 2]
Summer Term Examination

Class: Third Year.	Subject: Machine Design	Subject Code: PCC-ME308
Date: 20/06/2025	Time: 10.00-13.00	Max. Marks: 100

NB:

- This paper contains **02** pages (**09** Questions). All questions are compulsory.
- Outline your solution in clear-sequential steps.** Assume suitable data wherever necessary & clearly mention it.

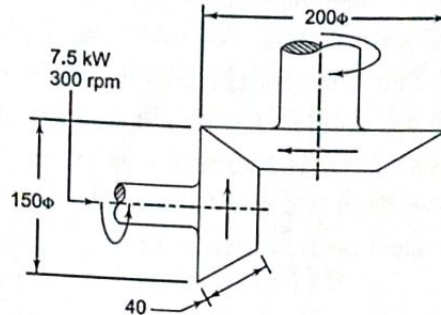
Q.No.	Questions	Marks	CO	BT
1.	It is required to design a chain drive to connect a 10 kW, 1400 rpm electric motor to a centrifugal pump running at 700 rpm. The service conditions involve moderate shocks, for which the <i>service factor</i> can be taken as 1.3. To reduce the polygonal effect, the number of teeth on the driving sprocket is to be selected as 17, so the <i>tooth correction factor</i> will be 1. Assume that the chain is a <i>simple roller chain</i> with only one strand (<i>Multiple strand factor</i> = 1). The ideal center-distance for chain drive is <i>forty times</i> the pitch of the chain. A. Select a proper roller chain from power rating catalogue and give a list of its dimensions. B. Determine the pitch circle diameters of the driving and driven sprockets. C. Determine the number of chain links required.	10	CO2, 4	BT3, 4
2.	The layout of a flat belt drive is shown in Fig. The mass of the belt is 2 kg per metre length and the coefficient of friction is 0.35. The belt is transmitting 35 kW of power. Calculate – A. the tensions on the tight and loose sides, and B. the length of the belt.	10	CO2, 3	BT3, 4
				
3.	A pair of straight teeth spur gears, having 20° involute full depth teeth is to transmit 22.5 kW when the pinion rotates at 300 RPM. The velocity ratio is 3 : 1. The ultimate tensile strength for the gear of cast iron and pinion of steel are 200 MPa and 300 MPa, respectively. The pinion should have 16 teeth, and the Face width is assumed to be 12 times the module. The service factor can be taken as 1. A minimum factor of safety = 1.5 is expected. Using <i>Lewis equation</i>, determine the <i>module</i>, <i>face width</i> and <i>pitch circle diameter</i> of gears. Velocity factor can be taken as, $C_v = \frac{6}{6+v}$ (v being the pitch line velocity in m/s.) And Lewis tooth form factor to be calculated using, $Y = 0.154 - \frac{0.912}{\text{No. of Teeth } (Z)}$	10	CO3, 4	BT3, 4
4.	A single-row deep groove ball bearing has a dynamic load capacity of 40500 N and operates on the following work cycle: i. radial load of 5000 N at 500 rpm for 25% of the time; ii. radial load of 10000 N at 700 rpm for 50% of the time; and iii. radial load of 7000 N at 400 rpm for the remaining 25% of the time. Calculate the <i>expected life</i> of the bearing in <i>hours</i>.	10	CO2, 4	BT3, 4

5. A helical compression spring, made of circular wire, is subjected to an axial force, which varies from 2.5 kN to 3.5 kN. Over this range of force, the deflection of the spring should be approximately 5 mm. The spring index can be taken as 5. The spring has square & ground ends. The spring is made of patented & cold-drawn steel wire with ultimate tensile strength of 1050 N/mm² and modulus of rigidity of 81370 N/mm². The permissible shear stress for the spring wire should be taken as 50% of the ultimate tensile strength. Design the spring and calculate – wire diameter; mean coil diameter; number of active coils; and solid length of the spring.

10 CO1, 2

6. A pair of bevel gears transmitting 7.5 kW at 300 rpm is shown in Fig. The pressure angle is 20°. Determine the components of the resultant gear tooth force.

10 CO1, BT1, 3 3



7. Derive Lewis Beam Strength Equation for spur gear tooth. Why is the pinion weaker than the gear made of same material?

10 CO3 BT2

8. You are assigned to design the motion & support system for a robotic arm used in assembly lines.

10 CO1, BT2, 2,4 3

- List suitable types of springs, bearings, and gears that could be used in such a system.
- Explain how each component contributes to motion transmission and load carrying.
- Select any two components and perform basic design calculations assuming reasonable values.

9. Write a short note on (Any Four) –

20 CO1, BT1, 4 2

- Flexible Drives used for power transmission
- Rolling Contact Bearing Failure – Causes & Remedies
- 'Self-Acting' Bearing (Hydrodynamic Journal Bearing)
- Importance of A.M. Wahl's factor in the design of helical springs
- 'Surge' in Helical Spring

Appendix A - Power rating of simple roller chain.

Pinion speed (rpm)	Power (kW)					
	10A	10B	12A	12B	16A	16B
700	5.91	6.71	10.73	11.63	23.26	27.73
1000	8.05	8.97	14.32	15.65	28.63	34.89
1400	11.18	11.67	14.32	18.15	18.49	38.47
1800	8.05	13.03	10.44	19.85	-	-
2000	7.16	13.49	8.50	20.57	-	-

- CO1 Student shows ability to design and model components using knowledge they acquired.
 CO2 Student exhibits ability to identify, formulate, and solves engineering problems.
 CO3 Student demonstrates the skill of designing motion transmission parts.
 CO4 Student contributes in design of new mechanical systems.

BT1 – Remember, BT2 – Understand, BT3 – Apply, BT4 – Analyze, BT5 – Evaluate, BT6 – Create.

Appendix B - Dimensions and breaking loads of simple roller chains.

ISO chain Number	Pitch p (mm)	Roller Diameter d _r (mm) (max)	Width b ₁ (mm) (min)	Transverse pitch p _t (mm)	Breaking load (min) (N)		
					Simple	Duplex	Triplex
10A (ANSI-50)	15.875	10.16	9.4	18.11	21 800	43 600	65400
10B	15.875	10.16	9.65	16.59	22 200	44 500	66700
12A (ANSI-60)	19.05	11.91	12.57	22.78	31 100	62 300	93400
12B	19.05	12.07	11.68	19.46	28 900	57 800	86700
16A (ANSI-80)	25.40	15.88	15.75	29.29	55 600	111 200	166800
16B	25.40	15.88	17.02	31.88	42 300	84 500	126800

	SGGS Institute of Engineering and Technology, Vishnupuri, Nanded		
	Academic Year: 2024-25	Semester: II	Examination: Summer Term
	Class: Third Year	Branch: Mechanical Engineering	
	Date: 21/06/2025	Time: 10:00 - 13:00	
Subject: Power Plant Engineering		Code: PEC-ME305	Maximum Marks: 100

General Instructions:

- I. This paper contains 1 (One) page. Figures to right indicate full marks.
- II. Draw neat & compact sketches wherever necessary.
- III. Assume data wherever required and state them clearly.

Que. No.	Question	Marks	CO	BT
1.	Explain different types of power generation plants in brief along with theoretical cycle on which it works.	10	CO1	BT1
2.	Explain different types of pulverised fuel burners in brief with schematic.	10	CO1	BT1
3.	List down any five site selection criterion for power plant, affecting its design, efficiency and economy.	10	CO1	BT1
4.	Explain the working principle and draw schematic of hydroelectric power plant. label it properly and write down their functions.	10	CO1	BT2
5.	Draught produced in chimney is 1.9 cm of water for the flue gases at temperature 290°C and ambient temperature 20°C. Flue gases to fuel ratio is 23 : 1. Assuming no losses and considering diameter of chimney as 1.8 m, calculate mass of flue gases flowing through the chimney.	10	CO2	BT5
6.	Following data refer to a four stroke double acting diesel engine having cylinder diameter 200 mm and piston stroke 350 mm. Calculate the mechanical efficiency of the engine. - m.e.p. on cover side = 6.5 bar - load on the break = 1370 N - m.e.p. on crank side = 7 bar - spring balance reading = 145 N - speed = 420 rpm - Brake wheel diameter = 1.2 m. - diameter of piston rod = 20 mm	15	CO3	BT5
7.	The air enters the compressor of an open cycle constant pressure gas turbine at a pressure of 1 bar and temperature of 20°C. The pressure of the air after compression is 4 bar. The isentropic efficiencies of the compressor and turbine are 80% and 85% respectively. The air fuel ratio used is 90:1. If flow rate of air is 3.0 kg/s, find: i. Power developed ii. Thermal efficiency of the cycle. Assume, $C_p = 1.0 \text{ KJ/KgK}$ and $\gamma = 1.4$ for air and gases. Calorific value of fuel = 41800 KJ/Kg.	15	CO1	B5
8.	Explain any ten terms associated with the economics of the power generation.	10	CO4	BT1
9.	List down the different pollutants of major concern from different power plants. Explain in brief how thermal pollution can be reduced in the thermal power plant.	5+5	CO2	BT2

**** ALL THE BEST ****

CO1: Student demonstrates working and constriction of various power plants.

CO2: Student exhibits knowledge of systems of waste disposals from power plant.

CO3: Student exhibits techniques of power generation and storage systems.

CO4: Student will be able to calculate the economy of power plant.

BT1: Remember, **BT2:** Understand, **BT3:** Apply, **BT4:** Analyze, **BT5:** Evaluate, **BT6:** Create.

	Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded Department of Mechanical Engineering Summer Term Examination Academic Year: 2024-25		
Class: TY	Subject: Manufacturing Processes - II	Subject Code: PCC-ME304	
Date: 19.06.2025	Time: 10:00 to 13:00	Max. Marks: 100	

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.

Q.N.	Solve all the questions	Marks	CO	BT
Q.1	Select the manufacturing processes for the given material and justify your answer with a scientific reason a) Alimina crucible with engraving b) Diamond cutter with cutting edges according to NRS system c) Stainless steel bearing outer ring	15	CO1 CO6	BT4
Q.2	How to convert tool angles from ASA to ORS through the graphical method and the analytical method?	15	CO2	BT2
Q.3	Explain in detail Broaching machine the principle, schematic diagram, types, advantages, disadvantages, and applications	15	CO4	BT1
Q.4	Elaborate Electro Discharge machine, Laser Beam Machining in detail with diagram, advantages, disadvantages and application	15	CO5	BT1
Q.5	During turning a steel rod of diameter 150 mm by a carbide tool of geometry; 0° , -12° , 8° , 6° , 15° , 60° , 0 (mm) at speed 560 rpm, feed 0.32 mm/rev. and depth of cut 4.0 mm the followings were observed $P_Z = 1000$ N, $P_Y = 200$ N, $a_2 = 0.8$ mm. Determine, without using MCD, the expected values of F , N , μ , P_s , P_n , τ_s , cutting power and specific energy requirement for the above mentioned machining operation.	10	CO1	BT3
Q.6	How much machining time will be required to reduce the diameter of a cast iron rod from 120 mm to 116 mm over a length of 100 mm by turning using a carbide insert. Reasonably select values of V_c and S_o .	10	CO3	BT3
Q.7	If in turning of a steel rod by a given cutting tool (material and geometry) at a given machining condition (s_o and t) under a given environment (cutting fluid application), the tool life decreases from	10	CO3	BT3

	80 min to 20 min. due to increase in cutting velocity, V_c from 60 m/min to 120 m/min., then at what cutting velocity the life of that tool under the same condition and environment will be 40 min.?			
Q.8	Material removal rate in AJM is $0.5 \text{ mm}^3/\text{s}$. Calculate material removal per impact if mass flow rate of abrasive is 3 gm/min, density is 3 gm/cc and grit size is $60 \mu\text{m}$ as well as indentation radius.	10	CO4	BT3

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Demonstrate the metal cutting process and select the appropriate cutting tools for it
CO 2	Exhibit the characteristic, terminologies related to cutting tools.
CO 3	Calculate the machining time, depth of cut, feed required.
CO 4	Demonstrate knowledge related to various machining processes.
CO 5	Exhibits the knowledge of various finishing processes.
CO 6	Select proper machining processes for the product to be manufacture